Reassessment of the microTEDAclus Clustering Algorithm and Proposal for an Improved Model

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Abstract

The abstract is deferred to after we finish the main content of the document. This allows us to write a more informed and concise summary of the work after all sections have been completed.

1 Introduction

The introduction is also deferred until we have a clearer picture of the entire document. We will revisit this section after we have completed the main body of the report to ensure it effectively sets the stage for our research and findings.

2 Background and Related Work

The TEDA framework is presented as an alternative to probability theory for handling data in non-deterministic scenarios [1]. The original microTEDAclus model applies its metrics to decide if a sample is an outlier to a microcluster and if microclusters are suitable candidates to form a macrocluster [4]. Finally, when evaluating the previous model's performance in the last-mile problem, [2] refactor the decay logic and variance computation.

2.1 Typicality and Eccentricity-Based Data Analytics

The paper *Outside the Box: An Alternative Data Analytics Frame-Work*, by Plamen Angelov, introduces the mathematical formulation for the metrics **Eccentricity** and **Typicality** [1].

Let

$$C = \{\sum_{i=1}^{k} \pi_i^k > 0, \quad j > 1, \quad k > 2\} \tag{1}$$

be a set of conditions for a sample j to be considered for the computation of eccentricity and typicality in a moment k.

Eccentricity, a measure of how much a sample deviates from a cluster, is defined as:

$$\xi_j^k = \frac{2\pi_j^k}{\sum_{i=1}^k \pi_i^k}, \quad C \tag{2}$$

Where π_j^k is the sum of distances from the sample j to all samples in the moment k.

Typicality, a measure of how well a sample fits within a cluster, is defined as:

$$\tau_j^k = 1 - \xi_j^k, \quad C \tag{3}$$

Both are, by definition, bounded: $0 < \xi_j^k < 1$ and $0 < \tau_j^k < 1$. The normalized eccentricity and typicality are presented as:

$$\zeta_j^k = \frac{\xi_j^k}{2}, \quad \sum_{i=1}^k \zeta_i^k = 1 \quad 0 < \zeta_j^k < \frac{1}{2} \quad C \tag{4}$$

$$t_j^k = \frac{\tau_j^k}{k-2}, \quad \sum_{i=1}^k t_i^k = 1 \quad 0 < t_j^k < \frac{1}{k-2} \quad C \tag{5}$$

The paper states that samples for which the normalized eccentricity $\zeta_j^k > 1/k$ are considered anomalous, and those for which the normalized typicality $t_j^k > 1/k$ are considered typical.

Furthermore, it provides recursive, or incremental, formulas for updating $\sum_{i=1}^k \pi_i^k$ in the squared Euclidean distance, as follows:

The mean and the scalar product of the samples in the moment k are

$$\mu^k = \frac{k-1}{k}\mu^{k-1} + \frac{1}{k}x_k \tag{6}$$

$$X^k = \frac{k-1}{k}X^{k-1} + \frac{1}{k}\|x_k\|^2 \tag{7}$$

From those,

$$\pi_j^k = k(\|x_j - \mu^k\|^2 + X^k - \|\mu^k\|^2) \tag{8}$$

and

$$\sum_{i=1}^k \pi_i^k = \sum_{i=1}^{k-1} \pi_i^{k-1} + 2\pi_k^k \tag{9}$$

Finally, it is made clear those metrics suit various proximity functions, among which are the cosine similarity and the Mahalanobis distance [3].

2.2 Original microTEDAclus

Evolving Clustering Algorithm Based on Mixture of Typicalities for Stream Data Mining, by Maia et al., employs the typicality and eccentricity metrics over the euclidean distance to instantiate and update microclusters from the detection of outliers and typical samples, as well as to decide when to merge microclusters into macroclusters [4].

It redefines the eccentricity expression into a recursive form:

$$\xi(x_k) = \frac{1}{k} + \frac{(\mu_k - x_k)^T (\mu_k - x_k)}{k\sigma_k^2} \tag{10}$$

where

$$\mu_k = \frac{k-1}{k}\mu_{k-1} + \frac{x_k}{k} \tag{11}$$

$$\sigma_k^2 = \frac{k-1}{k}\sigma_{k-1}^2 + \frac{1}{k-1}\|x_k - \mu_k\|^2 \tag{12}$$

Another change is in the criteria regarding the typicality and eccentricity thresholds. **microTEDAclus** considers a sample x_k to be an outlier if

$$\zeta(x_k) > \frac{m^2+1}{2k}, \quad m > 0 \tag{13}$$

In this algorithm, m is an arbitrary function of the number of samples in the data:

$$m(k) = \frac{3}{1 + e^{-0.007(k-100)}} \tag{14}$$

Microclusters are dynamically characterized by the following parameters:

- S_k^i: the number of samples in the microcluster i at moment k.
- μ_k^i: the mean of the samples in the microcluster i at moment k.
- σ_k^2: the variance of the samples in the microcluster i at moment k.
- $\xi^i(x_k)$, $\zeta^i(x_k)$: the eccentricity and normalized eccentricity of the sample x_k to the microcluster i at moment k.
- $\tau^i(x_k)$, $t^i(x_k)$: the typicality and normalized typicality of the sample x_k to the microcluster i at moment k.
- $D_k^i = 1/\zeta^i(x_k)$: the density of the microcluster i at moment k.
- $m_k^i(S_k^i)$: the dynamic threshold function for the microcluster i at moment k.

All the metrics and conditions yet defined for the entire data may be applied to each microcluster. Thus, for $S_k^i > 2$, a sample x_k is considered an outlier to the microcluster i if

$$\zeta^i(x_k) > \frac{m_k^i(S_k^i)^2 + 1}{2S_k^i} \tag{15}$$

and, for $S_k^i = 2$, if

$$\left(\zeta^i(x_k) > \frac{m_k^i(S_k^i)^2 + 1}{2S_k^i}\right) \wedge \left(\left(\sigma_2^i\right)^2 < r_0\right) \tag{16}$$

where r_0 is an hyperparameter of the model.

x_k is checked against all microclusters, which are updated if x_k is not an outlier to them as follows:

$$\begin{aligned} S_k^i &= S_{k-1}^i + 1 \\ \mu_k^i &= \frac{S_k^i - 1}{S_k^i}\mu_{S_k^i-1}^i + \frac{x_k}{S_k^i} \\ \left(\sigma_k^i\right)^2 &= \frac{S_k^i - 1}{S_k^i}\left(\sigma_{k-1}^i\right)^2 + \frac{1}{S_k^i - 1}\left(\frac{2\|x_k - u_k^i\|}{d}\right)^2 \\ \xi^i(x_k) &= \frac{1}{S_k^i} + \frac{2\left(\mu_k^i - x_k\right)^T\left(\mu_k^i - x_k\right)}{S_k^i\left(\sigma_k^i\right)^2 d} \end{aligned} \tag{17}$$

Macroclusters are the disjoint sets of connected microclusters in a graph where the vertices are the microclusters and the edges are defined by the condition

$$\text{dist}(\mu_k^i, \mu_k^j) < 2\left(\sigma_k^i + \sigma_k^j\right), \quad i \neq j \tag{18}$$

To prevent the collapse of all microclusters into a single macrocluster under overlap, a microcluster m_l^j is a candidate to form an edge only if

$$D_k^l \geq \text{mean}\left(D_k^l\right), \quad l = 1, \ldots, |\mathcal{M}_j| \tag{19}$$

where $\mathcal{M}_j = \left\{m_1^j, m_2^j, \cdots, m_l^j\right\}, \quad l = 1, 2, \cdots, N_j$ is the jth macrocluster composed of a set of N_j connected microclusters m_l^j.

The density of a macrocluster is the sum of the normalized typicalities of its active microclusters:

$$w_k^l = \frac{D_k^l}{\sum_{l \in \mathcal{M}_j} D_k^l} \tag{20}$$

Furthermore, the typicality of a sample x_k to a macrocluster $\mathcal{M}_j$ is the weighted sum of the typicalities of x_k to the microclusters in $\mathcal{M}_j$:

$$\mathcal{T}_j(x_k) = \sum_{l \in \mathcal{M}_j} w_k^l t_k^l(x_k) \tag{21}$$

A sample's predicted label j is the index of the macrocluster $\mathcal{M}_j$ to which it has the highest typicality:

$$j = \arg\max_j \mathcal{T}_j(x_k) \tag{22}$$

2.2.1 Critique

2.2.2 Order-Dependent Density as a Microcluster Characteristic

2.2.3 Anomalous Dimensionality Penalty

2.2.4 Unclear Microcluster Activation Criterion

2.3 microTEDAclus Updates in the *Mineração de Dados Aplicada a Problemas de Last-Mile na Logística* PHD Thesis

2.3.1 Critique

3 Reassessment

3.1 Data

3.2 Preprocessing

3.3 Processing

4 Proposal for an Improved Model

References

[1] Plamen Angelov. Outside the box: an alternative data analytics framework. *Journal of Automation, Mobile Robotics and Intelligent Systems*, pages 29–35, 2014.

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